



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

SPACE ODDITIES

EPFL COM-480 DATA VISUALIZATION
MILESTONE 2

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TABLE OF ACRONYMS

Acronym	Meaning
CDM	Conjunction Data Message
GEO	Geostationary Orbit
LEO	Low Earth Orbit
MEO	Medium Earth Orbit
MVP	Minimum Viable Product
NORAD ID	North American Aerospace Defense Command Identifier
SATCAT	Satellite Catalog
SGP4	Simplified General Perturbations Model 4
TCA	Time of Closest Approach
TLE	Two-Line Element set
UCS	Union of Concerned Scientists

TABLE 1
List of acronyms identified in the Space Oddities document.

Note: The symbol P_c is used in the text to denote the Probability of Collision.

1 PROJECT GOAL AND APPROACH

Space Oddities turns satellite catalogue data into a story that anyone can follow. The raw data (SATCAT catalogues, Two-Line Element sets, Conjunction Data Messages) is used daily by space agencies but is completely unreadable to a general audience.

The format is scrollytelling, inspired by The Pudding. In chapters 1 to 3, a single orbital map is pinned on the left while chapter cards scroll on the right, each filtering, recolouring, or labelling the same scene. In chapters 4 to 6, the left panel becomes domain-specific controls (a collision event list, then a CDM table) while the charts scroll alongside.

2 CHAPTERS AND KEY VISUALIZATIONS

2.1 CHAPTER 1: THE EXPONENTIAL GROWTH

Story: The space around Earth was once sparse, but has rapidly filled up, especially after 2019.

Visualization: A 2D Canvas orbital scene shows around 68 000 objects in orbit. Positions are computed from the latest available TLE using the SGP4 model via `satellite.js`, updated every 2 seconds to show continuous motion, with deorbited objects filtered out. A scroll-linked year slider progressively fills the map from 1957 to today. A D3 area/line chart shows yearly launches and the running total.

2.2 CHAPTER 2: WHERE DO THEY ORBIT?

Story: Satellites are not randomly distributed, but organized into distinct orbital regions dominated by different uses and actors. LEO is the most crowded region.

Visualization: The same orbital scene is recoloured by orbit band (LEO/MEO/GEO). A D3 bar chart shows object counts per band, and hovering a bar highlights only the corresponding satellites in the map.

2.3 CHAPTER 3: PAYLOADS VS. DEBRIS

Story: A large share of what orbits Earth is no longer active satellites, but debris.

Visualization: The orbital scene remains the main view, with type filter buttons to hide or show object categories. A D3 donut chart shows the payload/debris/rocket-body split, while selecting a category updates the map accordingly.

2.4 CHAPTER 4: KESSLER SYNDROME

Story: In orbit, one collision can create debris that increases the probability of future collisions.

Visualization: A list of real historical events is linked to the map, with hover-to-zoom on the corresponding case. A D3 debris-count line chart shows the evolution of fragmentation, with a filter on minimum object size (> 1 mm to > 1 m).

2.5 CHAPTER 5: VISUALIZING THE INVISIBLE RISK

Story: Collision risk is hard to grasp because near misses happen at extreme speed and usually remain invisible in raw tables.

Visualization: A filterable and sortable sticky CDM table lists 3 761 conjunction events. Picking a row reveals two satellite info cards, a 2D orbit-pair canvas, a time slider, and a D3 approach-distance graph. This turns abstract CDM values into a concrete close-approach visualization.

2.6 CHAPTER 6: THE MARGIN OF ERROR

Story: Satellite trajectories are uncertain, and this uncertainty is part of what makes close approaches dangerous.

Visualization: The CDM table remains fixed while the selected event is explored further. At TCA, 2D covariance ellipses are drawn around the two objects, the overlap region is shaded, and P_c is labelled. This makes the margin of error visible instead of an abstract probability.

EXTRA FEATURES

These features are optional additions on top of the MVP and will be implemented if time allows.

- **Kessler Cascade Simulator (Ch. 4):** Canvas animation of a ring of orbiting objects where a triggered collision causes fragments to spread and hit other objects.
- **Car Highway Comparison (Ch. 5):** SVG of two cars on a road with the gap scaled to the selected CDM miss distance and a time-to-contact label at orbital speed (7 km/s).
- **3D Margin of Error (Ch. 6):** Three.js view of uncertainty ellipsoids around TCA, rotatable by the user.

3 DATA, TOOLS AND LECTURES

Data: Space-Track.org SATCAT and TLE (around 68 000 objects); UCS Satellite Database (operator and owner metadata, joined on NORAD ID); 3 761 emergency-reportable CDMs covering 1 046 unique satellite pairs.

Tools:

- **React + Vite:** Component-based frontend; each chapter, sticky panel, CDM table, and satellite cards are separate components.
- **D3.js:** All statistical charts (area, bar, donut, line) and canvas quadtree hit-testing.
- **Canvas 2D API:** Orbital scene, orbit pair canvas, covariance ellipses, and the Kessler simulator (extra).
- **satellite.js (SGP4):** In-browser propagation; the approach distance graph computes positions at 2-minute intervals on demand.
- **Three.js / WebGL:** 3D covariance view (extra feature, Ch. 6).
- **Mantine:** UI primitives (badges, dividers).
- **Python (pandas, numpy):** Offline merge of SATCAT, UCS, and TLE files into `objects.csv`, `tle.csv`, `conjunction_events.csv`.

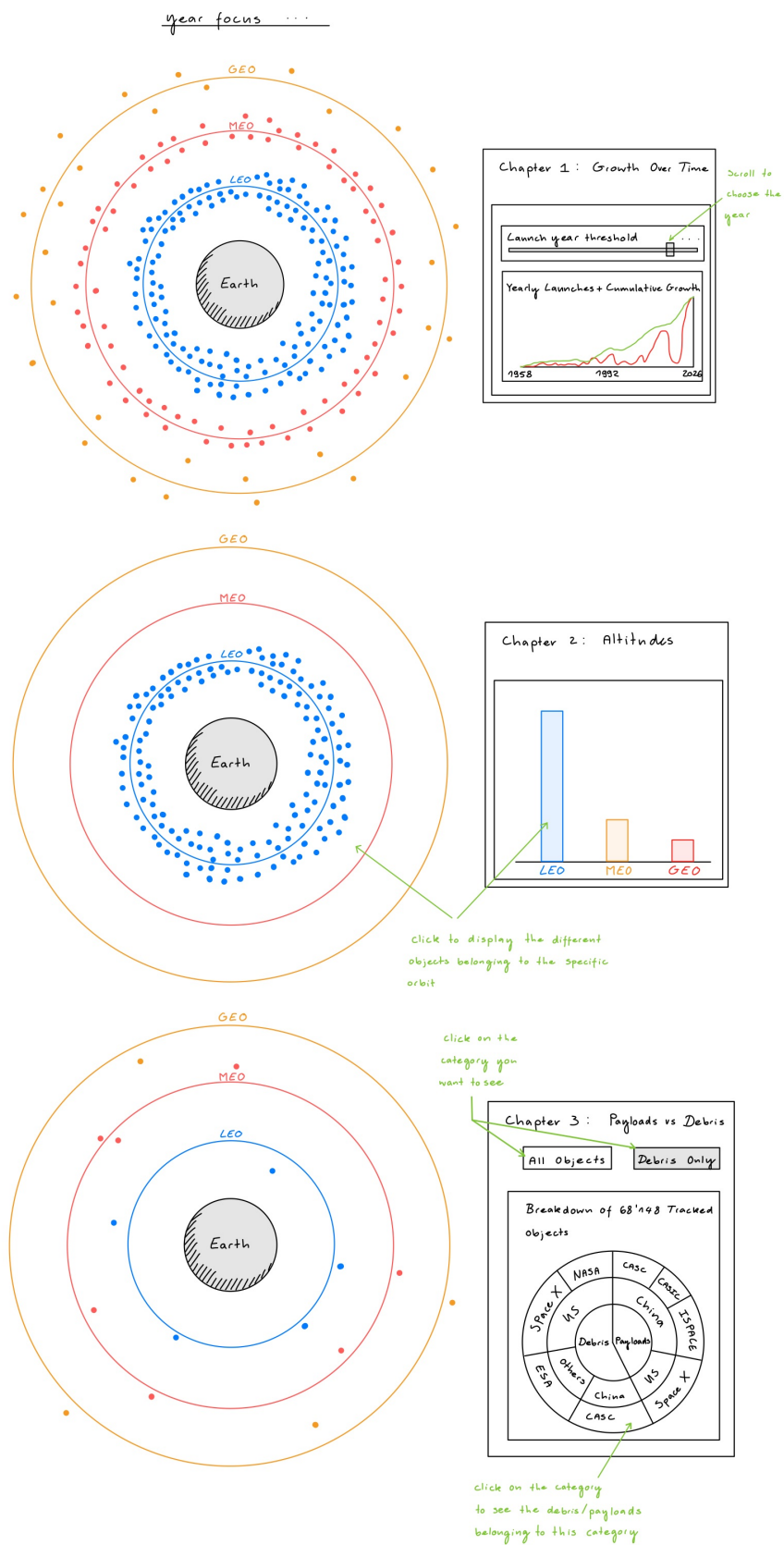
Relevant COM-480 Lectures:

- **Storytelling:** To structure the project as a "scrollytelling" experience with clear chapters and a guided narrative, as the website user is scrolling down, the different chapters unravel in a linear way.
- **Interaction, Views:** For scroll-linked transitions, filtering, and linked views between the orbital map, charts, and tables. For example, in Chapter 2, hovering a bar in the orbit distribution chart highlights only the corresponding satellites in the orbital map)
- **Perception, Color:** To help design readable color schemes and manage visual attention in a very dense scene, as the orbital map has thousands of objects. It should be easy to distinguish categories, for example in Chapter 2 where orbit types (LEO, MEO, GEO) are encoded with clearly separated colors.

- **Tabular Data:** To get inspiration on how to represent data effectively through charts (line, bar, donut) linked to the orbital map. For example, in Chapter 1, the yearly launches are summarized in a line/area chart.
- **D3.js:** Used to build charts and handle interactions between views (e.g. updating the bar chart and highlighting the corresponding satellites on the map on hover).
- **JavaScript, DOM:** To manage the page structure and interactions (scroll events, dynamic updates, data loading), especially for linking the scrollytelling flow with the different visual components.

4 WEBSITE LINK

Website link



Chapters

1

2

3

